

2018-Ph-H1-Q18

Section: Our Dynamic Universe

Topic: Relativity

A spacecraft travels at $0.8c$. According to an observer on Earth, a clock on the spacecraft runs:

- A. Fast
- B. Slow
- C. At the same rate
- D. At double speed

Solution:

Due to **time dilation**, moving clocks tick **more slowly** when viewed from Earth.

Answer: B

Revision Tip:

Remember the phrase: *Moving clocks run slow*. Time appears to slow down for objects moving close to the speed of light (from the observer's frame).